

UGED1111 Logic

Natural Deduction – Conditional Proof and Indirect Proof

Translation

Simple statement/ proposition: P, Q, R,

Logical Operators : $\sim$, $\cdot$, $\vee$, $\supset$, $\equiv$

Example: All cats are cute.

Categorical Syllogism: All cats are cute animals.

Propositional Logic: P

Translation

1. Snow is white. (**P**)
2. The Show is not white. ($\sim \mathbf{P}$)
3. Both *Ironman* and *Ant-man* are Marvel Characters. ($A \cdot B$)
 - *Ironman* is a Marvel Character. (A)
 - *Ant-man* is a Marvel Character. (B)
4. Winnie are clever or beautiful. ($M \vee N$)
 - Winnie are clever. (M)
 - Winnie are Beautiful. (N)
5. If Joe feels bored, he read books about logic. ($Y \supset W$)
6. Joe feels bored iff he read books about logic. ($Y \equiv W$)
 - Joe feels bored. (Y)
 - Joe read books about logic. (W)

Properties of Proposition

Tautologous

- **Necessarily true**
- **e.g. $P \vee \sim P$**
- e.g. Mimi is a cat or Mimi is not a cat.

Contingent

e.g. Mimi is a cat

Self-contradictory

- **Necessarily false**
- **e.g. $P \cdot \sim P$**
- e.g. Mimi is a cat and Mimi is not a cat.

Example

1. $N \supset O$

2. $N \supset P$

// $N \supset (O \cdot P)$

Conditional Proof



Conditional proof is a method for deriving a **conditional statement** (either the conclusion or some intermediate line) that offers a shorter and simpler way to derive the conclusion (comparing to direct method).

****Also**, some arguments have conclusions that **cannot** be derived by the direct method.
That's why we need it.

Conditional Proof



1. Assume the antecedent of the required conditional statement on one line (ACP step)
2. Derive the consequent on a subsequent line
3. “Discharge” this sequence of lines in a conditional statement that exactly replicates the targeted statement

****You can make more than one assumption**

Example

1. $N \supset O$

2. $N \supset P$

// $N \supset (O \bullet P)$

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Conditional Proof



Why can we make an assumption in the proof?

Do we assert a statement in doing the conditional proof?

NO!

Not asserting an statement as long as we discharge the assumption.

Discussion

$$1. A \supset (B \bullet C)$$

$$2. (B \vee D) \supset E \quad // A \supset E$$

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Discussion

1. $(G \vee H) \supset (S \bullet T)$

2. $(T \vee U) \supset (C \bullet D) \quad // \quad G \supset C$

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Example

1. $C \supset (A \bullet D)$

2. $B \supset (A \bullet E)$

// $(C \vee B) \supset A$

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Indirect Proof

Indirect proof can be used on any argument to derive either the conclusion or some intermediate line leading to the conclusion.

1. Assume the negation of the statement to be obtained.
2. Derive a contradiction from the assumption
3. Conclude that the original assumption is false
4. i.e. the targeted statement is true

** Sometimes you may need to use DN to derive the targeted statement

Example

1. $(S \vee T) \supset \sim S$

// $\sim S$

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Example

1. $\sim M \supset (N \bullet O)$

2. $N \supset P$

3. $O \supset \sim P$ // M

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Example

1. K

// $S \supset S$

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Remarks



After discharging the assumption, you cannot use any formula derived from that assumption to further derive.

Exercise

1. $A \supset \sim B$

2. $\sim(C \cdot \sim A)$ $// C \supset \sim B$

Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.